

Onur Doğan

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WORK EXPERIENCE

Machine Learning Engineer

June 2025 – Present

Lifemote

Design and develop ML solutions for telecommunications analytics, collaborating directly with executive and sales team to translate business needs into production-ready AI systems.

- Built and deployed **multimodal AI solutions** for device type identification on AWS, developing LLM-agent and embedding-based approaches over textual device metadata and time-series Wi-Fi behavior.
- Developed and deployed a production ML-based traffic classification model, using SQL-based aggregation logic to convert model predictions into subscriber profiles for large-scale ISP analytics; built a **full-stack** React/TypeScript dashboard for feedback collection and continuous model evaluation.
- Contributed to an anomaly explanation **LLM agent** that performs web search and generates explanations for analytical chart anomalies, while improving the LLM inference structure.
- Conducted research on transformer-based foundation models for Wi-Fi station time-series data, using PySpark and Scala for **large-scale preprocessing** in self-supervised representation learning workflows.

Computer Vision Engineer

December 2024 – June 2025

Arioviz, Germany – Part Time

- Developed a **real-time segmentation** model based on FastInst and SAM2 for robotic perception systems for a project of Ambyon, a Netherlands-based startup.
- Built data pipelines for training and evaluating robotic perception systems, using MongoDB, CVAT, and FiftyOne for data management, annotation workflows, and model evaluation.

Junior Data Scientist

Sept 2023 – March 2024

John Snow Labs, USA

Contributed to the development of the **open-source** NLU library. Also, contributed to Spark NLP for Healthcare projects and model integration on AWS and Databricks for large-scale PySpark workflows.

Junior Machine Learning Engineer

March 2023 – June 2023

GrainFox, Canada

- Developed and deployed an API for automatic information extraction from contract documents using computer vision techniques, reducing manual effort by 90% through automation.
- Integrated service with job queue into the production pipeline and contributed to MLOps implementations using FastAPI, Django, Docker, RabbitMQ, Celery, and Traefik, leading to a 35% reduction in latency.
- Developed an LLM-based document extraction agent using LangChain and the OpenAI API for processing unstructured documents.

SKILLS

Programming:	Python, Java, SQL, Scala, JavaScript/TypeScript, C++, C#, Git, Linux
Deep Learning / CV:	PyTorch, Hugging Face, Scikit-learn, PyG, vLLM, OpenCV, CVAT, FiftyOne
LLMs & Agents:	LangGraph, Pydantic-AI, AWS Bedrock, LangChain, LangSmith
MLOps / Full-Stack:	FastAPI, Django, Docker, React, RabbitMQ, Celery, PostgreSQL, MongoDB, Redis, Streamlit
Cloud / Big Data:	AWS, Databricks, PySpark, AWS EMR, AWS Glue

EDUCATION

MSc in Computer Science – Sabanci University

2024 – 2026

GPA: 3.61/4.0

- **Master’s Thesis:** CoFine: Hierarchy-Aware Semi-Supervised Learning for Fine-Grained Behavioral Classification
- **Advisors:** Prof. Oznur Tastan, Sabanci University; Mehmet Keles, Johns Hopkins University.
- **Relevant Coursework:** Scalable Learning Systems, Machine Learning, Deep Learning, Natural Language Processing, Network Science, Big Data Processing, IoT.
- **Teaching Assistant:** Programming Languages course, supporting 400+ students across two semesters.

BSc in Computer Engineering – Antalya Bilim University

2019 – 2024

GPA: 3.71/4.0

RESEARCH EXPERIENCE

Graduate Research Assistant

Jan 2025 – Present

Asst. Prof. Oznur Tastan – Sabanci University

Conducted research on **hierarchy-aware semi-supervised learning** and NLP applications in computational biology, focusing on sequential models, **protein language models**, and interpretability in NLP and LLMs.

Undergraduate Research Assistant

March 2022 – Nov 2023

Prof. Hilal Kazan – Antalya Bilim University

Contributed to **SciTuna**, an integration tool that uses network alignment to integrate multiple scRNA-seq datasets. Conducted research on differential gene co-expression networks.

Undergraduate Research Assistant

Sept 2020 – July 2022

Statistical Signal Processing Lab – Antalya Bilim University

Researched network inference on time-series action-potential data using Conditional Transfer Entropy and the IDTxL library.

SELECTED PUBLICATIONS

- A. Houdjedj, Y. Marouf, M. Myradov, **S. O. Doğan** et al., “SCITUNA: single-cell data integration tool using network alignment,” *BMC Bioinformatics*, 2025.
- M. U. Kahraman, **S. O. Doğan** et al., “Harmony in Diversity: Exploring Eclectic Chair Designs through Deep Neural Networks Analysis in Interior Design Education,” *New Design Ideas*, 2025.
- Y. Marouf, **S. O. Doğan** et al., “Sex-biased Expression of Neuroimmune Guidance Cues in Cardiovascular Diseases,” *presented at ERA-CVD and HIBIT 2022*

PROJECTS

Graph Neural Networks for Money Laundering Detection

[Link to Repo](#)

This project developed in the scope of Scalable Learning System Course and supervised by Associate Professor Kubilay Atasu

Domain Generalization for Surface Anomaly Detection

[Link to Repo](#)

Worked on domain generalization methods for detecting anomalies in different material types used in the industry.